

When Linear Attention Becomes a Memory-Update Problem

Standard Transformer attention gives a model direct access to previous tokens, but its inference memory grows with the context because the model retains key-value information from earlier tokens. Linear attention takes a different approach: it can compress past information into a recurrent state and update that state as new tokens arrive. Gated Linear Attention (GLA) makes this connection explicit: linear attention can be formulated as an RNN with a matrix-valued hidden state, enabling linear-time inference. [1]

This equivalence changes how we should think about efficient attention. The main challenge is not simply reducing computation; it is deciding **what information a compact state should retain, forget, and modify**. GLA introduces data-dependent gates that control the state update. In its experiments, a GLA Transformer trained on 2K-token sequences generalized beyond 20K tokens without significant perplexity degradation, showing that selective state updates can support long-context processing. [1]

But a compact state creates a fundamental trade-off. Arora et al. show that efficient sequence models with fixed-size recurrent states can struggle with **recall**: retrieving information that appeared earlier in the context. Their BASED model combines linear attention with sliding-window attention, allowing the effective state size to be adjusted. At up to 1.3B parameters, BASED matched strong sub-quadratic models in perplexity and improved real-world recall-intensive tasks by 10.36 accuracy points over the compared models. The authors also report 24× higher language-generation throughput than FlashAttention-2 when generating 1024 tokens with 1.3B-parameter models in their specified setup. [2]

This result points to a deeper issue: perhaps the important question is not only **how large the state is**, but **how it is updated**. DeltaNet addresses this by replacing the additive update used in standard linear attention with a delta-rule update. Instead of continually accumulating information, the update can make a more targeted modification to the stored state. The authors developed a hardware-efficient training algorithm and trained a 1.3B-parameter DeltaNet model on 100B tokens. In their experiments, it outperformed Mamba and GLA in perplexity and zero-shot downstream performance. [3]

Gated DeltaNet pushes this idea further by combining two forms of memory control. Gating provides adaptive memory erasure, while the delta rule enables targeted modifications. The authors describe these mechanisms as complementary and report that Gated DeltaNet consistently outperformed Mamba2 and DeltaNet across language modeling, common-sense reasoning, in-context retrieval, length extrapolation, and long-context understanding benchmarks. [4] The progression is therefore not simply a

sequence of new architectures: each step responds to a limitation exposed by the previous one.

The same problem remains active in 2026. Mamba-3 introduces a more expressive recurrence, complex-valued state updates, and a multi-input, multi-output formulation to improve the capabilities of efficient state-based models. At 1.5B parameters, it reports a 0.6-percentage-point improvement in average downstream accuracy over Gated DeltaNet; its MIMO variant adds another 1.2 points, for a total reported gain of 1.8 points. It also achieves comparable perplexity to Mamba-2 with half the state size in the reported state-size experiments. [5]

The limitation, however, should not be hidden: compressing a long history into a bounded state can make exact recall and state tracking difficult. BASED demonstrates the recall–state-size trade-off, while Mamba-3 itself identifies state tracking as a remaining weakness of many linear models. [2,5] More expressive updates can improve capability, but they also make the state dynamics more sophisticated.

BDH provides a related but distinct perspective. It is not a Linear Attention model. Instead, BDH is a biologically inspired attention-based state-space architecture whose inference-time working memory relies on synaptic plasticity and Hebbian learning. [6] Its relevance to this discussion is conceptual: both approaches make **persistent internal state and its modification** central to sequence processing, although they implement that idea very differently.

Taken together, the research supports a specific conclusion: **the recurrent formulation of linear attention exposes a memory-update problem at the heart of efficient sequence modeling. Once information is compressed into a bounded state, performance depends critically on deciding what to retain, erase, and modify.** Gating, delta updates, and increasingly expressive recurrences are not separate tricks; they are successive responses to that underlying constraint.

References

- [1] Yang et al. (2024) — Gated Linear Attention Transformers with Hardware-Efficient Training
- [2] Arora et al. (2024) — Simple Linear Attention Language Models Balance the Recall-Throughput Tradeoff
- [3] Yang et al. (2024) — Parallelizing Linear Transformers with the Delta Rule over Sequence Length
- [4] Yang et al. (2025) — Gated Delta Networks: Improving Mamba2 with Delta Rule
- [5] Lahoti et al. (2026) — Mamba-3: Improved Sequence Modeling using State Space Principles
- [6] Kosowski et al. (2025) — The Dragon Hatchling: The Missing Link between the Transformer and Models of the Brain